

Enhanced RF-Based Localization for Mobile Targets

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Abstract

Localization technology relying on Radio Frequency (RF) signals holds immense promise across intelligent environmental monitoring, security surveillance, and navigation domains. However, existing technologies often overlook the tilt and rotation states of targets during real-world movement. This limitation in accurately capturing target movement dynamics restricts the precision of localization. The existing data-driven methods for localization are highly dependent on the quantity and quality of data. But in practical scenarios, collecting sufficient data can be challenging due to costs or other constraints. This paper proposes a data enhancement strategy that generates a large amount of data suitable for complex scenarios by simulating RF signals under various environmental conditions with embedded physical constraints. Then, this paper designs a deep learning-based RF signal processing method that can learn the complex mapping relationship between RF signals and target states. By training the neural network, we can extract the tilt and rotation information of the target from the RF signals, thereby achieving high-precision estimation of the target's location. The experimental results show that the method proposed in this paper can achieve high-precision localization of moving targets, even in the case of changes in the tag's direction and interference from environmental noise.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**.

Keywords

Radio Frequency Identification, Localization, Data Augmentation, Dynamic Target Tracking

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1 Introduction

Location information is an important and indispensable information in modern industry[1–3]. Localization technology can continuously determine the position of a moving target, which attracts a lot of research works in the field of industrial information physical systems and is of great significance in practical scenarios[4, 5]. Localization techniques are broadly categorized into indoor and outdoor localization[6, 7]. For outdoor scenarios, satellite-based Global Positioning System (GPS) technology stands out for its high-precision location tracking capabilities. Conversely, in indoor scenarios, Radio Frequency Identification (RFID) technology has captured considerable interest from both academia and industry due to its cost-effectiveness, ease of deployment, and efficiency in data collection[8, 9]. The application of RFID extends to various domains such as indoor navigation, library inventory management, industrial manufacturing, and logistics and warehousing[10–12].

In recent years, researchers have pivoted towards utilizing RFID phase information for localization[13]. Capitalizing on the correlation between phase shifts and spatial distances, Tagoram[14] and MobiTagbot[15] have employed RF holograms to ascertain the likelihood of a tag's presence at a specific location. These systems have demonstrated the capability to achieve a centimeter-level accuracy, significantly advancing the field of RFID-based localization. The phase of an RFID tag is affected by both distance and orientation, with movement-induced tilts and rotations causing localization inaccuracies[16, 17]. Traditional methods, which assume constant orientation or treat tags as point masses, often overlook the orientation's substantial effect on phase changes. While some studies, like Tagyro[18], have explored the orientation's impact on phase by using tag arrays to infer object orientation from phase differences, they assume a static target position, an unrealistic condition in dynamic industrial settings. In industry, objects often change both position and orientation simultaneously. Ignoring orientation in localization models can lead to phase errors and reduced accuracy. Thus, incorporating orientation into RFID localization is essential for improving tracking reliability and precision in complex environments.

In this paper, we take into account the phase shifts due to tag rotation are investigated for full-angle localization. We propose an enhanced RF-based Localization (RFLoc) approach. The goal of this approach is to propose a full-angle localization technique that takes into account tag rotation, can be adapted to multiple scenarios, and can provide robust system performance. RFLoc faces the following two challenges that need to be addressed. First, how to construct

a localization model to generate trajectories and observations in a universal scenario. Second, how to make RFLoc applicable to various location-aware scenarios.

To address these questions, we first construct a tag distance model and a tag rotation model around the X-axis, Y-axis, and Z-axis, respectively. Secondly, simulated trajectories under various constraints are generated as training sets, and these trajectories can be converted into theoretical observations by the localization model. After that, the neural network technique is used to train multiple networks under different constraints, so that the trained networks can inversely sense the position information of the moving target according to the input real measurement data.

Our contributions are as follows. We construct the localization model, design different constraints, generate reasonable trajectories data sets, and convert these trajectories into theoretical observations. We develop a data-generation method that embeds physical constraints into the network to achieve data augmentation and reduce the cost of data collection.

2 System Design

2.1 State Space Model Construction

The phase and RSS information of RFID is not only related to the distance from the reader antenna to the tag, but is also affected by the rotation of the tag itself. If the tag is tilted or rotated during movement, the reader receives the sensed signal as a superposition of distance and rotation. Most of the existing localization work treats the tag as a point, or defaults to the tag does not change angle during the moving process, however, there are many moving and rotating scenarios in real life, such as production line conveyor belts, public transportation, and so on, which will cause the positioning deviation of the moving target in the actual application of the system.

The reader sends the RF signal to the tag via the antenna, which is then received by the tag's antenna. Since circular polarization can be achieved when the amplitudes of the horizontal and vertical components of the electric field are equal and the phase difference is $\frac{1}{2}\pi$ or $\frac{3}{2}\pi$, the transmitted signal of the reader, denoted as $S(t)$, can be represented as the superposition of cosine and sine signals in the horizontal and vertical directions:

$$S(t) = \mathbf{a} \cos(kt) + \mathbf{b} \sin(kt). \quad (1)$$

Here, $\mathbf{a}$ and $\mathbf{b}$ are the direction vectors of the reader antenna's horizontal and vertical linearly polarized antennas, respectively. The distance between the reader antenna and the tag is d . The transmitted signal $S(t)$ travels a distance of d to reach the tag and is projected onto the tag's polarization direction $\mathbf{c}$. The signal received by the tag, denoted as $S_t(t)$, can be expressed as:

$$S_t(t) = (\mathbf{a} \cdot \mathbf{c}) \cos(kt - \phi_d - \phi_t) + (\mathbf{b} \cdot \mathbf{c}) \sin(kt - \phi_d - \phi_t). \quad (2)$$

Here, $\phi_d = 2\pi d/\lambda$ is the phase shift caused by the distance d between the reader antenna and the tag, and ϕ_t is the phase shift caused by the tag's hardware.

The signal is emitted from the reader antenna, reaches the tag, and then is backscattered to return to the antenna. The returned signal can only be received along the two directions $\mathbf{a}$ and $\mathbf{b}$, which

are horizontal and vertical, respectively. The signals in these two directions are superimposed to form the received signal $R(t)$:

$$\left\{ \begin{array}{l} R(t) = R_a(t) + R_b(t) \\ = (\mathbf{a} \cdot \mathbf{c})^2 \cos(kt - 2\phi_d - \phi_t - \phi_r) + \\ (\mathbf{b} \cdot \mathbf{c})^2 \cos(kt - 2\phi_d - \phi_t - \phi_r) \\ = \cos(kt - 2\phi_d - \phi_t - \phi_r - \phi_0) \\ \tan \phi_0 = \frac{2(\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{c})}{(\mathbf{a} \cdot \mathbf{c})^2 - (\mathbf{b} \cdot \mathbf{c})^2} \end{array} \right. \quad (3)$$

Here, ϕ_d , ϕ_t , ϕ_r , and ϕ_0 are the phase shifts introduced by the tag-reader distance, the tag's hardware, the reader's hardware, and the tag's rotation, respectively.

The phase and RSS are generated using distance and rotation models, respectively. The position sequence is generated based on the 6D pose of the tag. The 6D pose refers to the position and orientation in three-dimensional space, typically represented by six degrees of freedom, namely three translational degrees of freedom and three rotational degrees of freedom. The translational degrees of freedom indicate the position of the object in the 3D coordinate system, denoted by the position vector $[x, y, z]$; the rotational degrees of freedom indicate the orientation of the tag in 3D space, here represented by the direction vector $[a, b, c]$ to indicate the angles of rotation around the X, Y, and Z axes.

2.2 Constraint-Embedded Data Generation

To reduce the time-consuming task of collecting large amounts of data on-site, a large amount of trajectory data that conforms to reality is generated based on the model. In this paper, a state generator is designed to automatically produce a series of states. First, many states subject to corresponding constraints are generated under different parameters. For example, different accelerations and velocities, as well as different starting points, can be simulated to produce tens of thousands of trajectories within just a few seconds. Then, predicted values are calculated based on the theoretical model to generate simulation sequences. Finally, the neural network is trained using the state dataset and the observation dataset. In this way, when reasonable data pairs appear in the training dataset, the network is forced to search for results within the constrained space, which is equivalent to embedding the constraints into the neural network.

To simulate the real movement of a moving target, this paper generates reasonable trajectories by embedding constraints, so that the network can make correct decisions. These constraints include movement and path constraints: Movement constraints refer to the movement patterns conforming to the laws of the physical world; Path constraints refer to the strict training of the movement trajectories to improve accuracy.

2.3 Collaborative Localization

Data augmentation is employed to set parameters for various scenarios to generate trajectories, eliminating the need for extensive manual data collection for the training set. We use the typical Long Short-Term Memory (LSTM) network to process the generated dataset. The training data input consists of approximately 10 dimensions, including the 6D pose coordinates of the tag, distance,

phase, RSS, and initial device parameters. Based on experimental experience, the network layer selects 64 hidden state dimensions. The number of hidden units determines the capacity and memory of the model. Larger values, such as 256, can also be chosen. The hidden states of the LSTM network need to be updated at each time step. Therefore, longer sequence lengths lead to more updates and longer training times, which can easily cause gradient vanishing and gradient explosion problems. In this case, the sequence length is set to 6 seconds according to the actual movement trajectory time length. Due to the large amount of data, to balance training accuracy and training time, this chapter sets the minimum training batch size to 2048 data points and the maximum training epoch to 8. The system simulates around 250,000 training scenarios, with random movement and rotation speeds, random movement ranges, and other parameters being fixed. The learning rate is an important hyperparameter in neural network training, determining the extent of weight updates in each iteration. A low learning rate leads to slow convergence, while a high learning rate causes unstable convergence. In this chapter, the learning rate is set to 0.05. According to experimental results, the LSTM network basically converges with the data from the second epoch.

During the testing phase, the measured values collected in the real environment are used as input, and the corresponding state data is obtained through the neural network. For each state, we can obtain a predicted sequence based on the model and corresponding parameters. To determine which candidate parameter is the best, we can calculate the reconstruction confidence. The desired parameter can make the predicted value consistent with the observed value. Finally, the confidence between the predicted value and the measured value measures the reliability of the network model.

3 Experiments

3.1 Implementation

The system employs a commercial off-the-shelf (COTS) UHF Imp-inJ Speedway R420 reader to implement the proposed system. The reader is compatible with the EPC Gen2 standard and reports the backscattered RF signals from the tags. The transmission power is set to 30dBm. The frequency band is 920.625 MHz – 924.875 MHz. Here, the frequency is set to 920.625 MHz to maintain stable phase and RSS readings. A circularly polarized Laird S9028PCR antenna is used, which provides reception and transmission of signals in the 902 MHz – 928 MHz band. The antenna gain is 9 dBic, and the 3 dB beamwidth is 70°. A Dell Inspiron 7560 laptop equipped with an Intel i77500U CPU and 16 GB RAM is employed. Experiments were conducted in open space and different indoor environments of a laboratory to verify the system performance.

The linear and circular trajectories are implemented using a cart track, as shown in Figure 1. We conduct experiments on two types of linear tracks in the horizontal X -axis direction and the vertical Y -axis direction (the line connecting the antenna and the tag). The figure displays the linear track in the Y -axis direction.

The circular trajectory is realized by deploying a circular track with a radius of 0.55 m, and the tag is carried by the cart and moves in a counterclockwise direction. In this chapter, the tag is placed on a toy cart, and the tag rotation is achieved by a rotating gimbal, which is placed on top of the cart. To evaluate the performance of

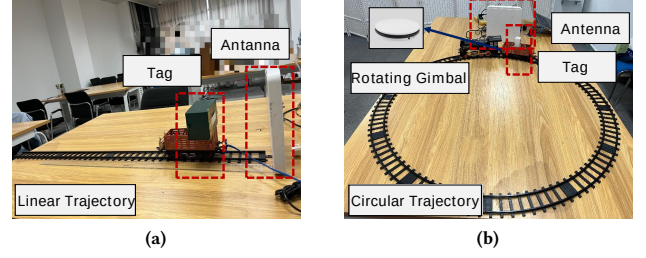


Figure 1: (a) Linear trajectory. (b) Circular trajectory.

position tracking, two scenarios are analyzed here: non-rotation tracking when the tag is tilted at a certain angle but does not rotate during movement, and rotation tracking when the tag rotates while moving. In the first case, the change in tag direction is caused only by movement, while in the second case, the change in tag direction is caused by both movement and tag rotation. The rotation speed of the rotating gimbal can be manually set, and here it is set to $10^\circ/\text{s}$.

3.2 Localization Performance without Rotation

Experiments are conducted in both horizontal and vertical linear track scenarios as well as circular track scenarios, with the tag moving at the same speed, which is set to 0.1 m/s. Tagoram[14] is a representative work of physical methods for mobile trajectory tracking. It achieves high-precision real-time localization and tracking of mobile tags by using a virtual antenna array based on differential enhanced holograms and analyzes both linear and circular track tracking. Therefore, this paper compares with it. The experimental results are shown in Figure 2. It can be seen from the experimental results that the location performance of MDLoc is superior to that of the model-based Tagoram scheme. When the tag moves along the horizontal X -axis track, the median localization errors of MDLoc and the model-based method are 2.58 cm and 5.07 cm, respectively. The 80% CDF cumulative errors are 3.8 cm and 4.3 cm, respectively. When the tag moves along the vertical Y -axis track, the median localization errors of MDLoc and the model-based method are 1.34 cm and 4.78 cm, respectively. The 80% CDF cumulative errors are 2.9 cm and 5.6 cm, respectively. When the tag moves along the circular C -axis track, the median localization errors of MDLoc and the model-based method are 1.97 cm and 2.51 cm, respectively. The 80% CDF cumulative errors are 3.0 cm and 4.5 cm, respectively.

3.3 Localization Performance with Rotation

This section verifies the situation where the tag rotates while moving. The moving speed is set to 0.1 m/s, and the tag rotation angular speed is set to $10^\circ/\text{s}$. The CDF of the localization error for the horizontal, vertical linear tracks, and the circular track is shown in Figure 3. When the tag rotates, it can be seen from the experimental results that the median localization error of RFLoc is better than that of the model-based method. The median localization errors are 4.3 cm and 7.7 cm, respectively. When the tag moves along the horizontal X -axis track, the median localization errors of RFLoc and the model-based method are 1.68 cm and 4.25 cm, respectively. When the tag moves along the vertical Y -axis track, the

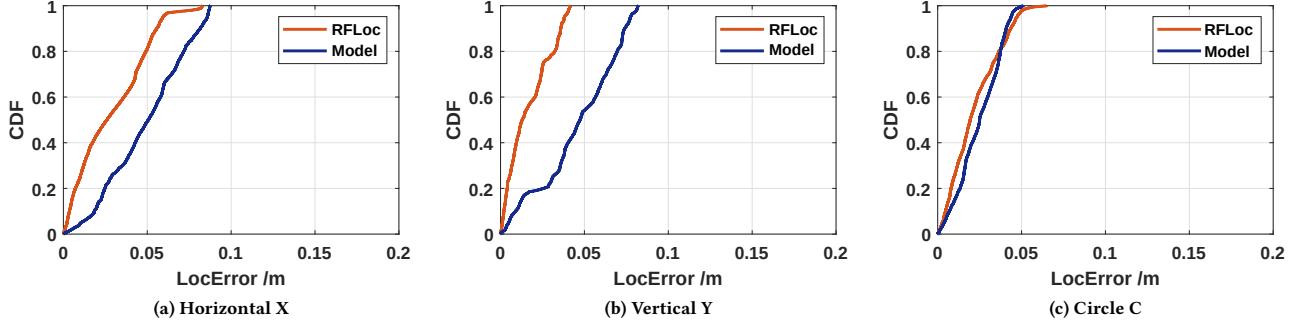


Figure 2: Localization performance evaluation without tag rotation.

median localization errors of RFLoc and the model-based method are 3.54 cm and 7.87 cm, respectively. When the tag moves along the circular C-axis track, the median localization errors of RFLoc and the model-based method are 2 cm and 2.5 cm, respectively. This is mainly because RFLoc constructs the mapping relationship between tag rotation and phase and RSS readings, and uses the tag rotation model to eliminate the localization errors introduced by tag rotation. Compared with the non-rotation tracking and rotation tracking scenarios, the experimental results show that RFLoc has stable localization accuracy. Regardless of whether the tag rotates or not, RFLoc can achieve robust localization perception. Compared with the localization tracking error of the model-based method, the experimental results show that the introduction of signal errors due to rotation has led to a certain degree of localization accuracy degradation.

4 Related Work

4.1 Physical-Driven Method

Localization is realized by establishing the mathematical relationship between perception data and target states [19, 20]. Physical-driven methods do not require training and testing work and can achieve localization [21], thus providing a detailed guide for understanding how moving target affects RF signals theoretically. This is based on a thorough understanding of the theoretical mathematical relationship between the physical space and the perceptual space. [22–24]. The core of this method is to extract physical quantities such as angle of arrival, time of flight, velocity, distance, Doppler shift, and frequency from the wireless propagation model to further achieve localization.

In recent years, some researchers have employed physical-driven methods. OmniTrack [10] enables mathematical analysis of how tag distance and orientation impact RF signals by constructing a phase model. This overcomes the limitation of conventional localization methods in tracking targets moving in non-single directions, achieving cm-level localization accuracy across diverse and universal scenarios. MRL [12] achieved 3D localization perception by constructing a mobility model. SiLoc [11] realized speed-independent localization by modeling the mobile scenario with an antenna array.

Physical-driven methods can infer target states by establishing the mathematical relationship between target states and signals.

These methods can clearly explain the relationship between various phenomena and targets, revealing the underlying patterns and mechanisms in the data. However, their limitation lies in the fact that for complex target states, specialized equipment or complex deployment is required, and it is difficult to establish a one-to-one mapping relationship in complex scenarios.

4.2 Data-Driven Method

Data-driven methods, also known as pattern-based or AI-based methods [25, 26], focus on discovering signal features to achieve localization. Initially, signal profiles need to be constructed during the training or learning phase. Subsequently, the signal features are extracted from the signals. Finally, localization is achieved through data-based approaches.

Data-driven methods rely on signal features, meaning that the same features correspond to specific state. If each state has a unique feature, data-driven methods have the potential to achieve localization [5]. Data-driven methods excel at accurately identifying complex state data through dataset analysis. They can adapt to diverse needs by learning and optimizing based on specific requirements and application scenarios. These methods also allow for performance scalability, integration of additional data sources, and adaptation to evolving demands and environments [27–29].

The limitations lie in the fact that data-driven methods are reliant on the training dataset. The quality or sample size of the dataset directly impacts the system's accuracy and stability. In this paper, we propose a data enhancement strategy that generates a large amount of data suitable for complex scenarios by simulating RF signals under different environmental conditions.

5 Conclusion

This paper considered tag rotation and developed an all-angle mobile RF localization system, RFLoc. It established a mathematical model linking distance, rotation, and RF information by analyzing how tag movement and rotation affect RF signals. A novel all-angle localization method driven by both physical models and data is proposed. By embedding physical constraints, it performed data augmentation across different speeds and environments, expands the training dataset, and cuts the cost of manual data collection.

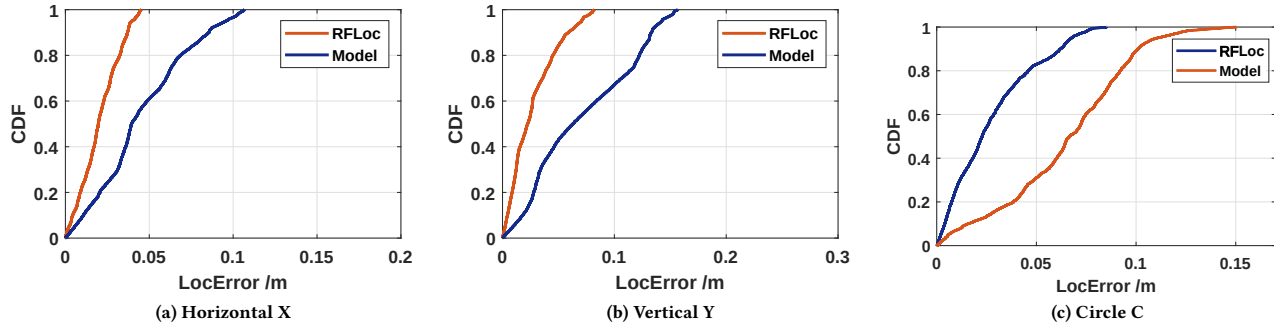


Figure 3: Localization performance evaluation with tag rotation.

Finally, an LSTM-based approach is employed for all-angle localization. Experiments in real-world settings show that RFDLoc achieves high-precision localization under its current design.

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